# Aida Khaleghi

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## Research Interests

- **Computational Biology**
- **Single-Cell Perturbation Modeling**
- **Tissue Foundation Models**
- **Medical Imaging Analysis**

## Education

**M.Sc. in Artificial Intelligence and Robotics** Sep 2024 – Mar 2027 (expected)
*Iran University of Science and Technology, Tehran, Iran*

- Thesis: "Deep Unfolding-Based Anomaly Detection in Medical Imaging" (in progress)
- Supervisors: Prof. Behrouz Minaei-Bidgoli (Google Scholar) and Dr. Maryam Amoozegar (Google Scholar)
- GPA: 19.05/20 (4.0/4.0)

**B.Sc. in Computer Engineering** Sep 2020 – Jul 2024
*Shahid Bahonar University of Kerman, Kerman, Iran*

- Bachelor project: "A Heterogeneous Graph Representation of Text via Graph Neural Networks for Persian Sentiment Analysis" (grade: 20/20)
- Supervisor: Prof. Mahdi Eftekhari (Google Scholar)
- GPA: 19.04/20 (3.98/4.0)

## Research Experience

**Research Assistant** Mar 2026 – Present
*Kerman Graduate University of Technology, Kerman, Iran*
*Under supervision of Dr. Maryam Amoozegar*

- **D**esigned a hybrid network built on IFISTA-ROTAB, an online tensor-based background subtraction method that separates each frame into a low-rank background and a sparse foreground
- Replaced the update of the temporal-dependency vector with a ConvLSTM and the foreground update with a convolutional network, while keeping the recursive least squares (RLS) updates of the background subspace factors
- Implemented and trained the network as sole technical contributor, fully online and unsupervised: the model processes one frame at a time and trains without ground-truth masks; redesigned the loss function to provide a stronger training signal
- Improved F-measure by at least 0.1 over IFISTA-ROTAB on LASIESTA and CDNet2014 datasets

**Graduate Researcher** Jan 2025 – Present
*Data Mining Lab, Iran University of Science and Technology, Tehran, Iran*
*Under supervision of Prof. Behrouz Minaei-Bidgoli and Dr. Maryam Amoozegar*

- Designing a deep unfolding network for anomaly detection in medical imaging modalities like MRI and CT which preserves the interpretability of the underlying iterative solver unlike deep learning networks
- Leading problem formulation, architecture design, PyTorch implementation, and evaluation on medical datasets (experiments in progress)
- Reviewed prompt tuning techniques for large language models for master seminar

**Undergraduate Researcher** Feb 2024 – Jul 2024
*Shahid Bahonar University of Kerman, Kerman, Iran*
*Under supervision of Prof. Mahdi Eftekhari*

- **A**dapted a heterogeneous graph neural network method for text classification, originally developed for English, to Persian in a team of 3 students

- Rebuilt the text-to-graph pipeline for Persian, using fastText word embeddings and Hazm for part-of-speech tagging and dependency parsing, to represent each review as a graph of word, sentence, tag, and dependency nodes
- Trained and evaluated a heterogeneous graph attention network on the Digikala and SnappFood review datasets

## Manuscripts

**A. Khaleghi**, M. Amoozegar. "Online Unsupervised Hybrid Network for Moving Object Segmentation in Videos." *Manuscript in preparation*, 2026.

**A**. Ghasemi, B. Minaei-Bidgoli, and **A. Khaleghi**, "Universal Noise Augmentation (UNA): A Language-Agnostic Framework for Robust Multilingual Text Classification," manuscript under review at [Machine Learning with Applications], 2026.

## Industry Experience

**AI Engineer** Aug 2025 – Dec 2025

*Vida Medicals, Canada (remote)*

- Improved breast asymmetry classification accuracy on mammography from 60% to 82% by replacing a hand-crafted feature pipeline with transfer learning, then adding a multi-task objective with auxiliary labels to handle data scarcity
- Developed breast tumor segmentation models using nnU-Net and YOLO, reaching 0.83 Dice Score on internal mammography dataset
- Built and deployed a Streamlit annotation tool use to label mammography images under a shared protocol by radiologists

## Awards & Honors

- Ranked 4th out of 20 students in the Artificial Intelligence & Robotics M.Sc. cohort – Iran University of Science and Technology, 2026
- Admitted to the M.Sc. program without entrance examination through the Exceptional Talent track, reserved for top-ranked undergraduates nationwide – Iran University of Science and Technology, 2024
- Ranked 1st out of +90 students in the Computer Engineering class – Shahid Bahonar University of Kerman, 2024

## Selected Projects

- **Perturbench: Predicting Unseen Double Gene Perturbations** — Benchmarked a CPA-style conditional VAE against additive and mean baselines for predicting unseen double gene perturbations in single-cell Perturb-seq data (Norman 2019), using validation-only model selection, 5 seeds, and bootstrap confidence intervals. The additive baseline remained best (MSE 0.045 vs. 0.057); analysis showed that the best CPA variant has a collapsed latent space and that its error rises sharply for test pairs dissimilar to training perturbations
- **Drug–Drug Interaction Prediction with GNNs** — Formulated drug–drug interaction prediction as link prediction on the ogbl-ddi graph, using a GraphSAGE encoder and an MLP pair scorer; compared BCE and AUC losses and tuned hyperparameters on validation data across multiple seeds. Reached 0.896 test Hits@20 on a 500-drug subset and 0.845 when transferred without re-tuning to 2,134 drugs
- **Multi-Task Breast Ultrasound Analysis** — Extended a single-task MONAI segmentation baseline into a multi-task U-Net that jointly segments breast tumors and classifies them as benign or malignant (BUS-BRA, 1,875 images), with a classification head on the shared bottleneck and patient-grouped splits. Reached 0.82 Dice and 0.84 ROC-AUC on a held-out test set
- **Instance Segmentation under Adverse Weather** — Generated a 3,704-frame instance segmentation dataset in the CARLA simulator across clear, rain, fog, and night conditions, auto-labelled with GroundedSAM and split by ego-vehicle position to test spatial generalization. Compared YOLOv11n-seg and YOLOv11m-seg, and achieved a 4.1× inference speedup (11.9 ms to 2.9 ms) with FP16 TensorRT

## Teaching Experience

**Graduate Teaching Assistant** Sep 2025 – Present

*Iran University of Science and Technology, Tehran, Iran*

- **Courses:** Natural Language Processing, Machine Learning, Neural Networks, Deep Learning (Fall 2026); Advanced Topics in Computer Vision (Spring 2025); Deep Learning, Applied Artificial Intelligence (for students outside computer department) (Fall 2026)
- Designed programming assignments and problem sets, and graded them with written feedback
- Mentored six graduate students on their final research projects, advising on paper selection, implementation, and evaluation; one Machine Learning course project grew into a co-authored paper (UNA) which now submitted (see Manuscripts)
- For the NLP course, delivered a hands-on workshop on parameter-efficient fine-tuning (PEFT), and designed, invigilated, and graded the practical exam

**Undergraduate Teaching Assistant** Jan 2023 – Aug 2024
*Shahid Bahonar University of Kerman, Kerman, Iran*

- **Courses:** Natural Language Processing (sole TA) (Spring 2024); Operating Systems (sole TA), Algorithms Design and Analysis (Spring 2023)
- Designed and graded assignments for all three courses
- Delivered a workshop on the Transformer architecture for the NLP course, and led midterm review and problem-solving sessions for Algorithms and Operating Systems

## Skills

- **Programming:** Python, C++, SQL, C#
- **Machine Learning:** PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph
- **Medical Imaging & Vision:** MONAI, nnU-Net, Ultralytics YOLO, OpenCV, Albumentations
- **Computational Biology:** Scanpy, AnnData, PyTorch Geometric
- **Scientific Computing:** NumPy, Pandas, Matplotlib, Seaborn
- **Tools & Deployment:** Git/GitHub, Docker, Linux/Bash, TensorRT, ONNX, FastAPI, PostgreSQL, Streamlit, TensorBoard, LaTeX, uv
- **Languages:** English (fluent), Persian (native)

## References

**Prof. Behrouz Minaei-Bidgoli**
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Iran University of Science and Technology
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*M.Sc. thesis supervisor*

**Dr. Maryam Amoozegar**
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*M.Sc. thesis co-supervisor and research collaborator*

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*Instructor of courses I assisted as TA*

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*Instructor of the NLP course I assisted as TA*